

In this generation, the use of plastic materials is considered a norm in our daily lives. They can be seen in various forms of daily products, including food containers, bottles, cutleries and single-use plastic bags. Among these, single-use plastic bags are one of the most used items which can contribute to the environmental pollution. It is undoubtedly important for us to revise our lifestyle and consider using other alternatives methods or products to reduce the usage of plastic materials in our daily lives. As a result, there has been a global movement to ban on single-use plastic bags to protect marine life, reduce ongoing environmental and safeguard humans' health.

First and foremost, many believe that the widespread use of single-use plastic bags should be prohibited globally to curb the harm inflicted on marine biodiversity. This is largely because marine animals who often mistake plastic debris for food, leading to the ingestion of plastics. Due to their size, shape, and colour, which often resemble plankton species (Ding et al., 2018), microplastics are frequently ingested by marine invertebrates and fish. (Ayassamy, 2025, p. 2). This phenomenon can cause fatal digestive blockages for the marine animals. In addition, massive plastic waste causes accumulation of trash on coastlines, which leads to the fatal entanglement of marine life. This increases the risk of marine animals becoming trapped in the waste when they accidentally swim into it, making them unable to free themselves. For instance, seals may become tangled in plastic bands, causing them to struggle to breathe and eventually die. Not only that, but plastic bags also eventually break down into microplastics through exposure to sunlight, waves, and weathering. These microplastics then scatter throughout the ocean and are ingested by small marine organisms. Larger marine animals then consume the contaminated small organisms, disrupting the marine ecosystem as the microplastics move up the marine food chain while carrying toxic chemicals. As a result, the global prohibition of single-use plastic bags is crucial to protect marine biodiversity, as it

would reduce plastic pollution, prevent the deaths of marine organisms, and help preserve the stability of ocean ecosystems for future generations.

Furthermore, the significant reason to prohibit plastic bags is their destructive impact on environment. Plastic bags contain chemical additives with varying level of toxicity, which can harm both human health and the environment. These substances have the potential to pollute the ground, air and water. Moreover, such pollutions occur through both the decomposition and the manufacturing of plastic bags. In terms of decomposition, plastic bags do not degrade naturally; instead, they take approximately around 20 to 500 years or more to degrade (Mentis et al., 2022, p. 2). Due to this reason, the accumulation of decomposing plastic bags in land and marine ecosystems leads to soil and water contamination, while their breakdown also emits harmful gases such as carbon dioxide and methane (Bowen Li et al., 2022, p. 4). Not only that, the manufacturing of the plastic bags also adds to serious environmental harm. This is because, the production of plastic bags relies on non-renewable natural resources such as natural gases and crude oil. The extraction process of these raw materials releases significant amounts of air pollutants into the atmosphere (Bowen Li et al., 2022, p. 1). Therefore, the combined effect of these processes is the release of vast amounts of toxic gases during plastic manufacturing and decomposition, which contribute to air pollution and add to the concentration of greenhouse gases, thus worsening global warming and climate change.

In addition to that, the use of plastic bags should be outlawed to ensure that humans' health is always protected from chronic diseases. This is because an excessive use of plastic bags will increase the risk of being exposed to the toxic materials in the plastic bags. As we all know, plastic bags are made of petroleum, which is a type of materials that contains heavy metal particles. These particles are often released in a significant amount when plastic bags are exposed to sunlight for a period of time (Karak, Parveen, Modak, Adhikari, & Chakraborty,

2025, pp. 11 - 12). These released particles will be transmitted into food sources through grocery plastic bags or into water supplies through the breaking down of plastic bags in the soils, which will then be absorbed into our bodies. This action could cause serious health issues such as irritations, inflammations or even damage to the internal organs due to high concentration of heavy metals in our bodies (Karak et al., 2025, p. 12). This happens because heavy metals, such as Zinc and Lead, can carry microorganisms and malicious pathogens from contaminated surfaces into our organs (Karak et al., 2025, p.12). This as a result will increase the risk of being infected by deadly diseases. For example, cancer, cardiovascular problems and respiratory issues. Therefore, to avoid these chronic diseases, we should reduce the use of plastic bags in our daily life as it will decrease the chances of being in contact with toxic particles.

Next, plastic bags also provide several benefits in human life, as they offer many advantages and are convenient for everyday use (Li et al., 2022, p. 1). When it comes to grocery shopping, people usually require bags that are practical and easy to carry because groceries often include heavy items. Plastic bags are lightweight yet capable of carrying heavy loads. Furthermore, the use of plastic bags in the food industry is very widespread because plastic is a durable and airtight material. This helps to maintain the freshness of food for a longer period of time. However, the use of plastic bags can still cause a great deal of damage in the long term. For example, in restaurants, hot food is often packed using plastic materials immediately after cooking. When plastic is exposed to high temperatures, its original chemical structure may become unstable. As a result, toxic chemicals such as BPA and phthalates, as well as particles like microplastics, may seep into the food. Consequently, this may lead to long-term health risks such as hormonal disruption, cancer, and other health problems (Li et al., 2022, p. 2). In addition, plastic bags that are not properly managed can also negatively affect animals. For instance, plastic bags thrown into the sea may be mistaken for jellyfish by turtles. When turtles

consume the plastic, it cannot be digested properly and creates a false sense of fullness. As a result, the turtles may stop eating and eventually die from starvation. Based on the examples given, it can be concluded that the use of plastic bags can cause significant harm not only to humans but also to animals and the environment (Li et al., 2022, pp. 3 - 4).

As a conclusion, banning the single use of plastic bags worldwide is a necessary step to reduce ecological damage. Even though plastic bags still provide benefits in everyday human life, the harm inflicted not only on human health, but also on marine biodiversity and the environment, has proven that these items bring far more negative consequences than benefits. These findings suggest that if the use of plastic bags continues to grow over time, it will increase pollution and lead to the destruction of natural habitats and biodiversity. For the future of our environment, urgent initiatives must be taken to prevent this crisis. For instance, the government can tighten regulations related to the use of plastic bags, such as limiting their widespread usage in supermarkets. Not only that, but they can also implement stricter waste management systems and promote more eco-friendly alternatives. Therefore, it is strongly hoped that the use of plastic bags can be reduced so that the beauty of the environment can continue to be enjoyed by future generations.

References

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